

Section 8 solution

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1 Exercises

1.1 True or False?

1. A normal good can be a Giffen good.

False. A Giffen good is a good whose positive income effect is greater in absolute value than its negative substitution effect. All goods with positive income effects are inferior goods.

2. Low-income people are more likely to take the bus than high-income people. That means that bus tickets are an inferior good.

True. In this example, bus tickets are a textbook example of an inferior good: demand for the good is decreasing in income.

3. If h^* is a Hicksian demand function, then $\partial h^*/\partial p$ isolates the income effect.

False. This derivative is the definition of the substitution effect, not the income effect.

4. A normal good x is one such that $\partial x^*/\partial M > 0$, so that means that the income effect for a normal good is positive.

False. The income effect is $(-x^*)(\partial x^*/\partial M)$. The first term is always negative, so if the second term is positive, then the income effect is negative. This is intuitive: I consume more of a normal good as I get richer and less of it as I get poorer. If the price of the good goes up, then I effectively get poorer, so I consume less of it, and therefore the income effect is negative.

5. Bread bought and sold among peasants during the French Revolution is a plausible candidate for a Giffen good.

True. Here is a plausible story: Peasants like to consume bread and meat, but bread is a necessity and meat is a luxury. When the price of bread goes up, peasants get poorer, so they can no longer afford any meat at all. They therefore put all their money into bread, so their bread consumption goes up even though bread's price goes up.

1.2 Slutsky Decomposition for Stone-Geary

1. Set up the expenditure-minimization problem for the utility function $u = x_2(x_1 + 4)$ and the constraint $u \geq U_0$. **Answer:**

$$\min_{(h_1, h_2) \in \mathbb{R}_+^2} p_1 h_1 + p_2 h_2$$

$$s.t. \ h_2(h_1 + 4) \geq U_0 \text{ and } h_i \geq 0 \text{ for each } i = 1, 2$$

2. Drop the nonnegativity constraints and impose $u = U_0$. Set up the Lagrangian. **Answer:** $L(h_1, h_2, \lambda; p_1, p_2, U_0) = p_1 h_1 + p_2 h_2 - \lambda(h_2(h_1 + 4) - U_0)$

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3. Find the FOCs. **Answer:** The first order conditions:

$$\begin{aligned}\frac{\partial L}{\partial \lambda} &= h_2(h_1 + 4) - U_0 = 0 \\ \frac{\partial L}{\partial h_1} &= p_1 - \lambda h_2 = 0 \\ \frac{\partial L}{\partial h_2} &= p_2 - \lambda(h_1 + 4) = 0\end{aligned}$$

4. Solve for h_1^* and h_2^* and assume an interior solution. **Answer:**

$$\begin{aligned}\frac{p_1}{p_2} &= \frac{h_2}{h_1 + 4} \\ \Rightarrow h_2 &= (h_1 + 4) \frac{p_1}{p_2}\end{aligned}$$

With the utility constraint:

$$\begin{aligned}(h_1 + 4) \frac{p_1}{p_2} (h_1 + 4) - U_0 &= 0 \\ h_1^* &= \sqrt{\frac{p_2}{p_1} U_0} - 4 \\ \Rightarrow h_2^* &= \left(\sqrt{\frac{p_2}{p_1} U_0} - 4 + 4 \right) \frac{p_1}{p_2} = \sqrt{\frac{p_1}{p_2} U_0}\end{aligned}$$

5. **Answer:**

$$\begin{aligned}x_1^* &= \frac{M - 4p_1}{2p_1} \\ x_2^* &= \frac{M + 4p_1}{2p_2}\end{aligned}$$

6. Are these goods normal or inferior? **Answer:**

$$\begin{aligned}\frac{\partial x_1^*}{\partial M} &= \frac{1}{2p_1} > 0 \\ \frac{\partial x_2^*}{\partial M} &= \frac{1}{2p_2} > 0\end{aligned}$$

Thus both goods are normal.

7. Decompose the $\partial x_1 / \partial p_1$ into a substitution effect and an income effect. **Answer:**

Substitution Effect:

$$\begin{aligned}\frac{\partial h_1^*(p_1, p_2, U_0)}{\partial p_1} \Big|_{U_0=v(p_1, p_2, M)} &= -\frac{1}{2} \frac{p_2 U_0}{\sqrt{\frac{p_2}{p_1} U_0}} \frac{1}{(p_1)^2} \Big|_{U_0=v(p_1, p_2, M)} \\ &= -\frac{p_2}{2(p_1)^2} \sqrt{\frac{p_1}{p_2} \frac{(M + 4p_1)^2}{4p_1 p_2}} \\ &= -\frac{M + 4p_1}{4(p_1)^2}\end{aligned}$$

Income Effect:

$$-x_1^* \frac{\partial x_1^*}{\partial M} = -\frac{M - 4p_1}{2p_1} \left(\frac{1}{2p_1} \right) = -\frac{M - 4p_1}{4p_1^2}$$

8. Now what about when $M < 4p_1$, in which case we have a corner solution? What are the Hicksian demands? What are the substitution and income effects? **Answer:**

- By inspection $x_2^* = \frac{M}{p_2}$ and $x_1^* = 0$.
- Can we use the earlier derived h_1^* and h_2^* ? No, because $M < 4p_1$ means that $h_1^* < 0$. Instead, we have to set $h_1^* = 0$ and then using our constraint $h_2(h_1 + 4) = U_0$ we solve for $h_2^* = U_0/4$. And that's it!
- From here, we can see that the **substitution effects** are zero ($\frac{\partial h_1^*}{\partial p_1} = \frac{\partial h_2^*}{\partial p_2} = 0$). As we change price of either good (on the margin) there is no substitution towards or away from the other good because the corner solution is still the optimum.
- However, there is an **income effect** for the change in p_2 of $-x_2^* \cdot \frac{\partial x_2^*}{\partial M} = -\frac{M}{p_2} \frac{1}{p_2} = -\frac{M}{p_2^2}$.

1.3 Slutsky Decomposition for Cobb-Douglas

Consider a consumer with Cobb-Douglas preferences:

$$u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$$

By assumption, $0 < \alpha < 1$. Recall that you already solved for the Marshallian demand functions (i.e., you already solved the *utility-maximization problem*) on your homework:

$$x_1^*(p_1, p_2, M) = \frac{\alpha M}{p_1}, \quad x_2^*(p_1, p_2, M) = \frac{(1-\alpha)M}{p_2}$$

Similarly, you already solved for the Hicksian demand functions (i.e., you already solved the *expenditure-minimization problem*) in lecture:

$$h_1^*(p_1, p_2, \bar{u}) = \bar{u} \left[\left(\frac{p_2}{p_1} \right) \left(\frac{\alpha}{1-\alpha} \right) \right]^{1-\alpha}, \quad h_2^*(p_1, p_2, \bar{u}) = \bar{u} \left[\left(\frac{p_1}{p_2} \right) \left(\frac{1-\alpha}{\alpha} \right) \right]^\alpha$$

1. Draw the **Engel curve** for x_1^* , meaning draw how x_1^* changes with M (holding prices constant). **Answer:** If we place x_1^* on our vertical axis and M on our horizontal axis, then it's easy to see that this is just an upward sloping line that intersects the origin and that has a slope of $\frac{\alpha}{p_1}$.
2. Derive the indirect utility function $v(p_1, p_2, M) = u(x_1^*(p_1, p_2, M), x_2^*(p_1, p_2, M))$. Verify that $x_i^*(p_1, p_2, M) = h_i^*(p_1, p_2, v(p_1, p_2, M))$ **Answer:** $v(p_1, p_2, M) = \left(\frac{\alpha M}{p_1} \right)^\alpha \left(\frac{(1-\alpha)M}{p_2} \right)^{1-\alpha} = \left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{(1-\alpha)}{p_2} \right)^{1-\alpha} M$. Verify for good 1. (The case for good 2 is symmetric.)

$$\begin{aligned} h_1^*(p_1, p_2, v(p_1, p_2, M)) &= \left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{(1-\alpha)}{p_2} \right)^{1-\alpha} M \left[\left(\frac{p_2}{p_1} \right) \left(\frac{\alpha}{1-\alpha} \right) \right]^{1-\alpha} \\ &= (\alpha)^{\alpha+(1-\alpha)} M = \frac{\alpha M}{p_1} \\ &= x_1^*(p_1, p_2, M) \end{aligned}$$

3. Derive the expenditure function $e(p_1, p_2, \bar{u}) = p_1 h_1^*(p_1, p_2, \bar{u}) + p_2 h_2^*(p_1, p_2, \bar{u})$. Verify that $x_i^*(p_1, p_2, e(p_1, p_2, \bar{u})) = h_i^*(p_1, p_2, \bar{u})$. **Answer:**

$$\begin{aligned} e(p_1, p_2, \bar{u}) &= p_1 \left(\frac{p_2}{p_1} \frac{\alpha}{1-\alpha} \right)^{1-\alpha} \bar{u} + p_2 \left(\frac{p_1}{p_2} \frac{1-\alpha}{\alpha} \right)^\alpha \bar{u} \\ &= \bar{u} \left[p_1^\alpha \left(\frac{p_2}{p_1} \frac{\alpha}{1-\alpha} \right)^{1-\alpha} + p_2^{1-\alpha} \left(\frac{p_1}{p_2} \frac{1-\alpha}{\alpha} \right)^\alpha \right] \\ &= \bar{u} p_1^\alpha p_2^{1-\alpha} \left[\left(\frac{\alpha}{1-\alpha} \right)^{1-\alpha} + \left(\frac{1-\alpha}{\alpha} \right)^\alpha \right] \end{aligned}$$

Verify for good 1. (The case for good 2 is symmetric.)

$$\begin{aligned}
x_1^*(p_1, p_2, e(p_1, p_2, \bar{u})) &= \frac{\alpha}{p_1} \left(\bar{u} p_1^\alpha p_2^{1-\alpha} \left[\left(\frac{\alpha}{1-\alpha} \right)^{1-\alpha} + \left(\frac{1-\alpha}{\alpha} \right)^\alpha \right] \right) \\
&= \bar{u} \left(\frac{p_2}{p_1} \right)^{1-\alpha} \left[\frac{\alpha^{2-\alpha}}{(1-\alpha)^{1-\alpha}} + \frac{(1-\alpha)^\alpha}{\alpha^{\alpha-1}} \right] \\
&= \bar{u} \left(\frac{p_2}{p_1} \right)^{1-\alpha} \left[\frac{\alpha + (1-\alpha)}{(1-\alpha)^{1-\alpha} \alpha^{\alpha-1}} \right] \\
&= \bar{u} \left(\frac{p_2}{p_1} \right)^{1-\alpha} \left(\frac{\alpha}{1-\alpha} \right)^{1-\alpha}
\end{aligned}$$

4. Decompose the $\partial x_1 / \partial p_1$ into a substitution effect and an income effect. **Answer:**
Substitution effect:

$$\begin{aligned}
\frac{\partial h_1^*}{\partial p_1} \Big|_{\bar{u}=v(p_1, p_2, M)} &= \frac{\partial}{\partial p_1} \left[\bar{u} \left[\left(\frac{p_2}{p_1} \right) \left(\frac{\alpha}{1-\alpha} \right) \right]^{1-\alpha} \right] \Big|_{\bar{u}=v(p_1, p_2, M)} \\
&= -\frac{(1-\alpha)}{p_1} \bar{u} \left[\left(\frac{p_2}{p_1} \right) \left(\frac{\alpha}{1-\alpha} \right) \right]^{1-\alpha} \Big|_{\bar{u}=v(p_1, p_2, M)} \\
&= -\frac{(1-\alpha)}{p_1} \left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{1-\alpha}{p_2} \right)^{1-\alpha} M \left[\left(\frac{p_2}{p_1} \right) \left(\frac{\alpha}{1-\alpha} \right) \right]^{1-\alpha} \\
&= -M p_1^{-(1+\alpha)-(1-\alpha)} p_2^{-(1-\alpha)+(1-\alpha)} \alpha^{\alpha+1-\alpha} (1-\alpha)^{1+(1-\alpha)-(1-\alpha)} \\
&= -\frac{\alpha(1-\alpha)M}{p_1^2}
\end{aligned}$$

Income effect:

$$\begin{aligned}
-x_1^* \frac{\partial x_1^*(p_1, p_2, M)}{\partial M} &= \left(\frac{\alpha M}{p_1} \right) \frac{\partial}{\partial M} \left(\frac{\alpha M}{p_1} \right) \\
&= -\left(\frac{\alpha}{p_1} \right)^2 M
\end{aligned}$$

5. Verify that the Slutsky equation holds for $\partial x_1 / \partial p_1$. See equation 3 above. **Answer:** Add the substitution effect and the income effect:

$$-\left(\frac{\alpha}{p_1} \right)^2 M - \frac{\alpha(1-\alpha)}{p_1^2} M = -\frac{\alpha M}{p_1^2}$$

Differentiate the Marshallian demand function with respect to p_1 :

$$\frac{\partial x_1^*(p_1, p_2, M)}{\partial p_1} = \frac{\partial}{\partial p_1} \left[\frac{\alpha M}{p_1} \right] = -\frac{\alpha M}{p_1^2}$$

They are the same!

6. Verify that the Slutsky equation holds for $\partial x_1 / \partial p_2$. See equation 4 above. **Answer:**

Substitution effect:

$$\begin{aligned}
\frac{\partial h_1^*}{\partial p_2} \Big|_{\bar{u}=v(p_1, p_2, M)} &= \frac{\partial}{\partial p_2} \left[\bar{u} \left[\left(\frac{p_2}{p_1} \right) \left(\frac{\alpha}{1-\alpha} \right) \right]^{1-\alpha} \right] \Big|_{\bar{u}=v(p_1, p_2, M)} \\
&= \frac{(1-\alpha)}{p_2} \bar{u} \left[\left(\frac{p_2}{p_1} \right) \left(\frac{\alpha}{1-\alpha} \right) \right]^{1-\alpha} \Big|_{\bar{u}=v(p_1, p_2, M)} \\
&= \frac{(1-\alpha)}{p_2} \left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{(1-\alpha)}{p_2} \right)^{1-\alpha} M \left[\left(\frac{p_2}{p_1} \right) \left(\frac{\alpha}{1-\alpha} \right) \right]^{1-\alpha} \\
&= M p_1^{-[\alpha+(1-\alpha)]} p_2^{-1-(1-\alpha)+(1-\alpha)} \alpha^{\alpha+1-\alpha} (1-\alpha)^{1+(1-\alpha)-(1-\alpha)} \\
&= \frac{\alpha(1-\alpha)M}{p_1 p_2}
\end{aligned}$$

Income effect:

$$\begin{aligned}
-x_2^* \frac{\partial x_1^*(p_1, p_2, M)}{\partial M} &= \left(\frac{(1-\alpha)M}{p_2} \right) \frac{\partial}{\partial M} \left(\frac{\alpha M}{p_1} \right) \\
&= - \left(\frac{\alpha(1-\alpha)}{p_1 p_2} \right)
\end{aligned}$$

Add the substitution effect and the income effect:

$$\frac{\alpha(1-\alpha)M}{p_1 p_2} - \left(\frac{\alpha(1-\alpha)M}{p_1 p_2} \right) = 0$$

Differentiate the Marshallian demand function with respect to p_2 :

$$\frac{\partial x_1^*(p_1, p_2, M)}{\partial p_2} = \frac{\partial}{\partial p_2} \left[\frac{\alpha M}{p_1} \right] = 0$$

They are the same!